Histograms with ggplot2

Let's go over how to create histograms with ggplot2. Refer to the video for the full explanation! Also a quick note, we are going to be showing a lot of what ggplot *can* do, but **not** what you *should* do!

Load Data

We'll use the movie dataset that comes with ggplot:

In [17]:

```
library(ggplot2)
df <- movies <- movies[sample(nrow(movies), 1000), ]
```

In [18]:

```
head(df)
```

Out[18]:

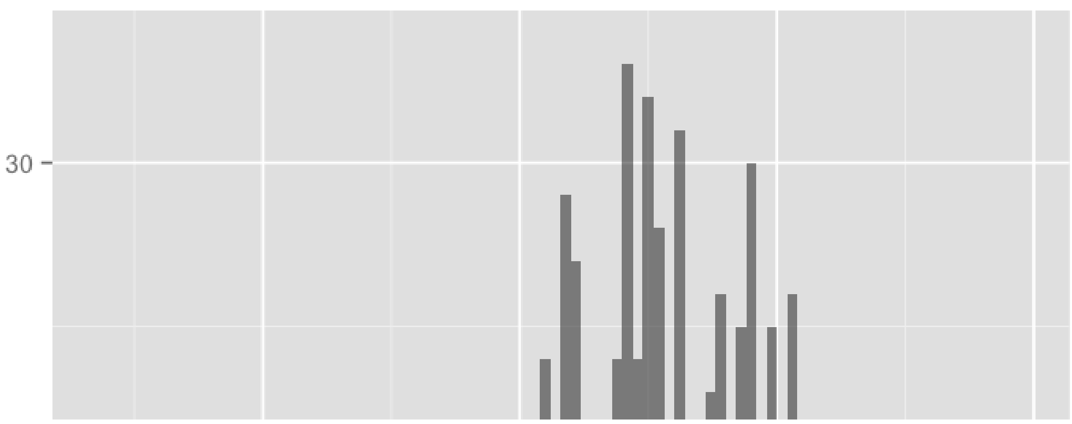
	title	year	length	budget	rating	votes	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10	mpaa	A
21569	Gunan il guerriero	1983	81	NA	2	38	44.5	14.5	4.5	14.5	14.5	0	4.5	4.5	4.5	4.5		1
39820	Phantom Brother	1988	92	NA	2	10	45.5	14.5	0	0	0	24.5	14.5	0	0	14.5		0
47662	Snow	1996	81	NA	7.2	10	0	0	14.5	0	0	44.5	14.5	24.5	0	24.5		0
41080	Prelude	2000	6	NA	6.3	18	4.5	0	0	14.5	4.5	0	24.5	24.5	4.5	24.5		0
58741	Zwaarmoedige verhalen voor bij de centrale verwarming	1975	95	NA	5	27	14.5	0	0	4.5	14.5	14.5	14.5	14.5	14.5	24.5		0
47430	Sleepy-Time Tom	1951	7	NA	6.7	32	0	0	0	14.5	4.5	4.5	24.5	14.5	24.5	4.5		0

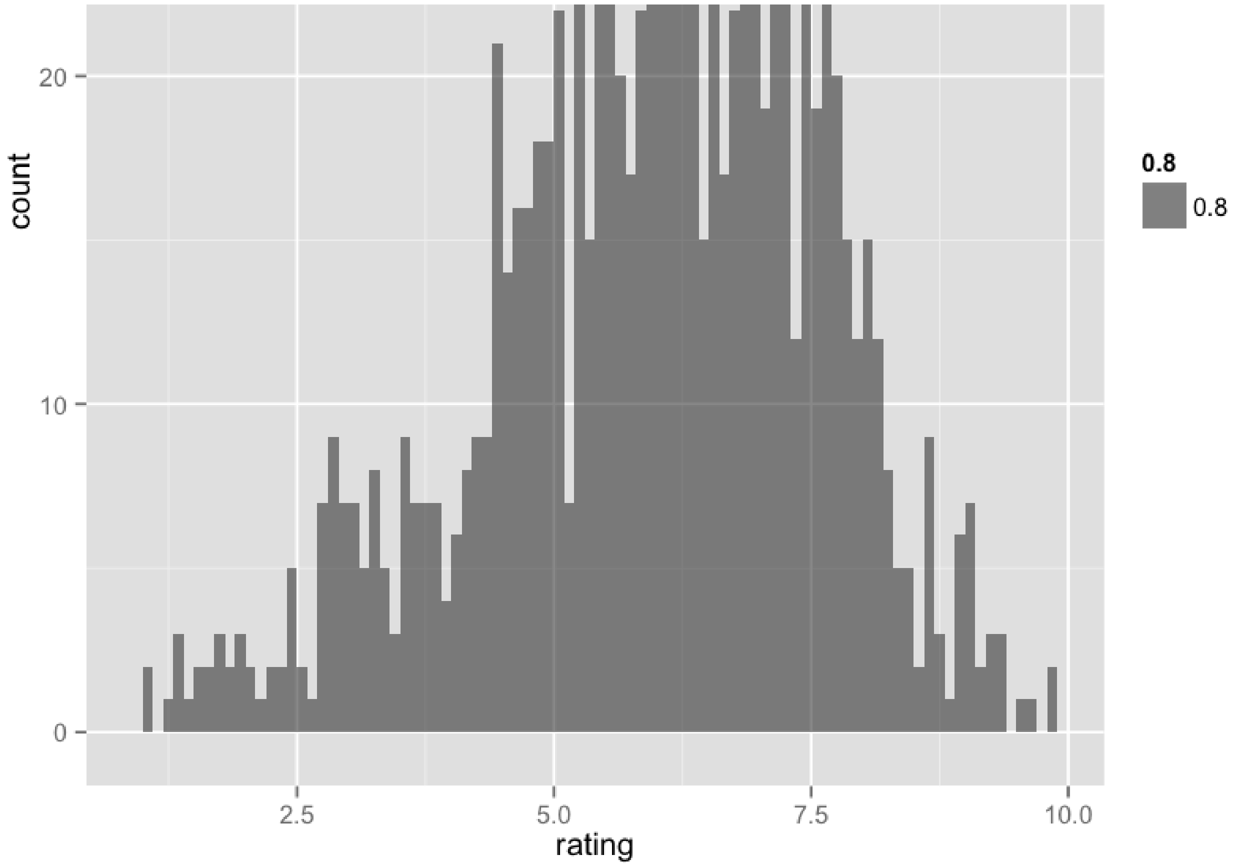
Using qplot()

Basics

In [21]:

```
qplot(rating,data=df,geom='histogram',binwidth=0.1,alpha=0.8)
```





Using ggplot()

Let's see how we can really expand on this by using ggplot! They syntax starts off with the base plot:

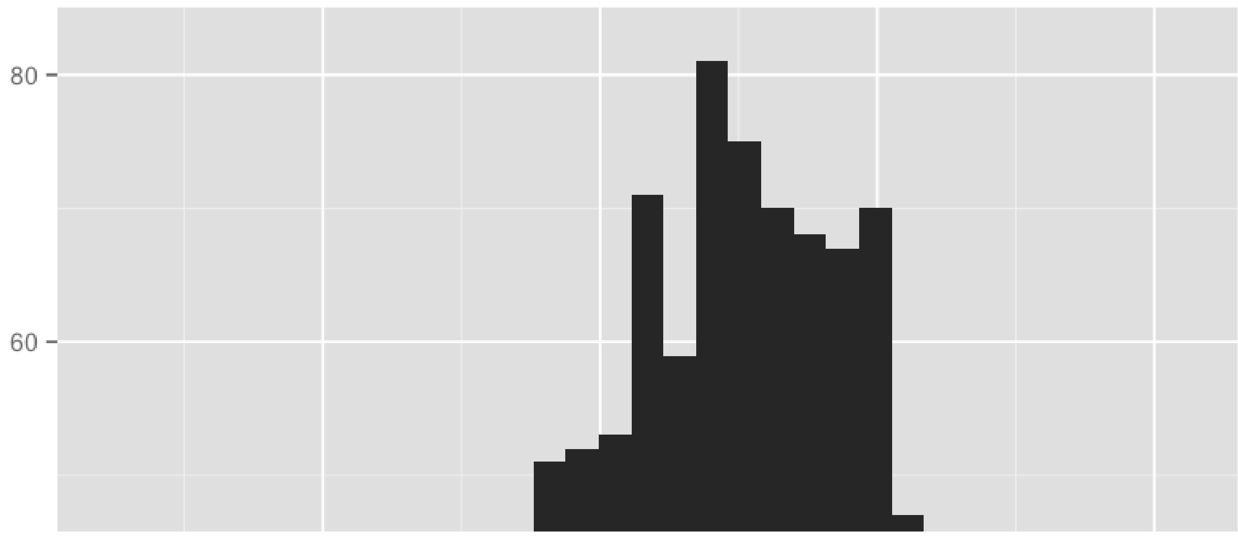
In [23]:

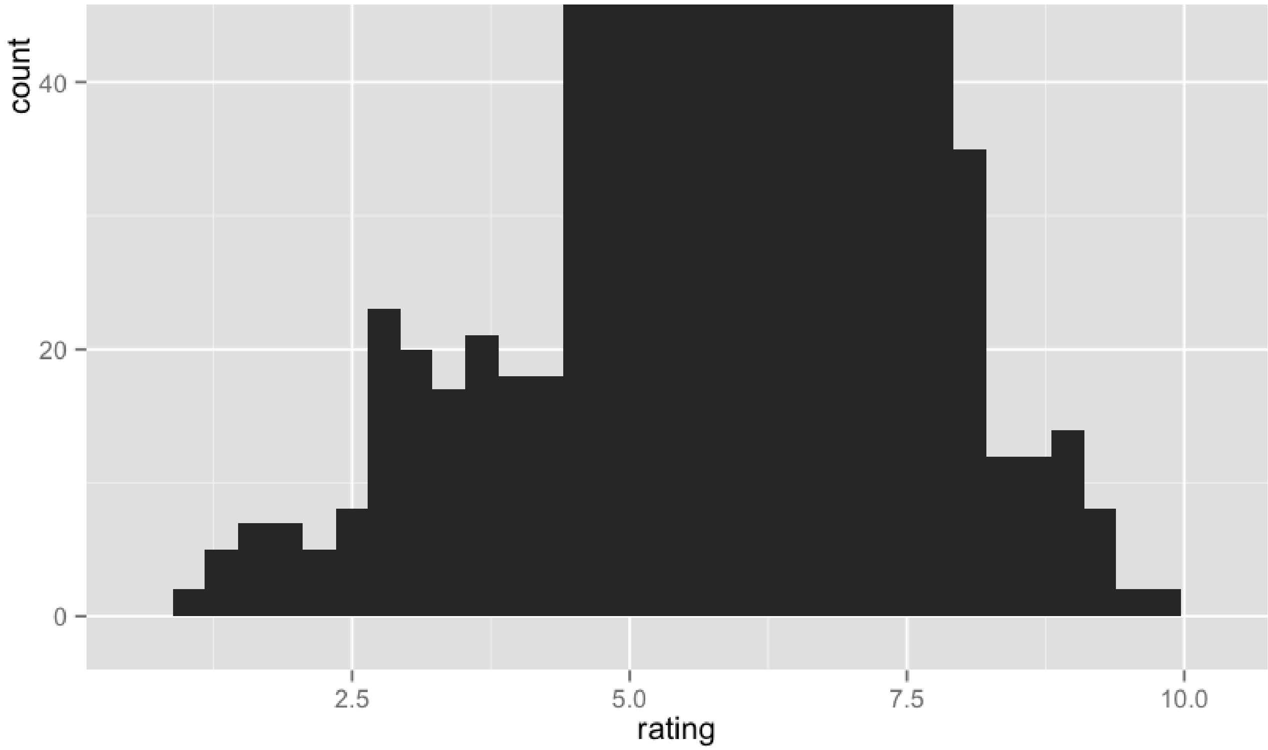
```
# ggplot(data, aesthetics)
pl <- ggplot(df,aes(x=rating))
```

In [24]:

```
# Add Histogram Geometry
pl + geom_histogram()
```

```
stat_bin: binwidth defaulted to range/30. Use 'binwidth = x' to adjust this.
```

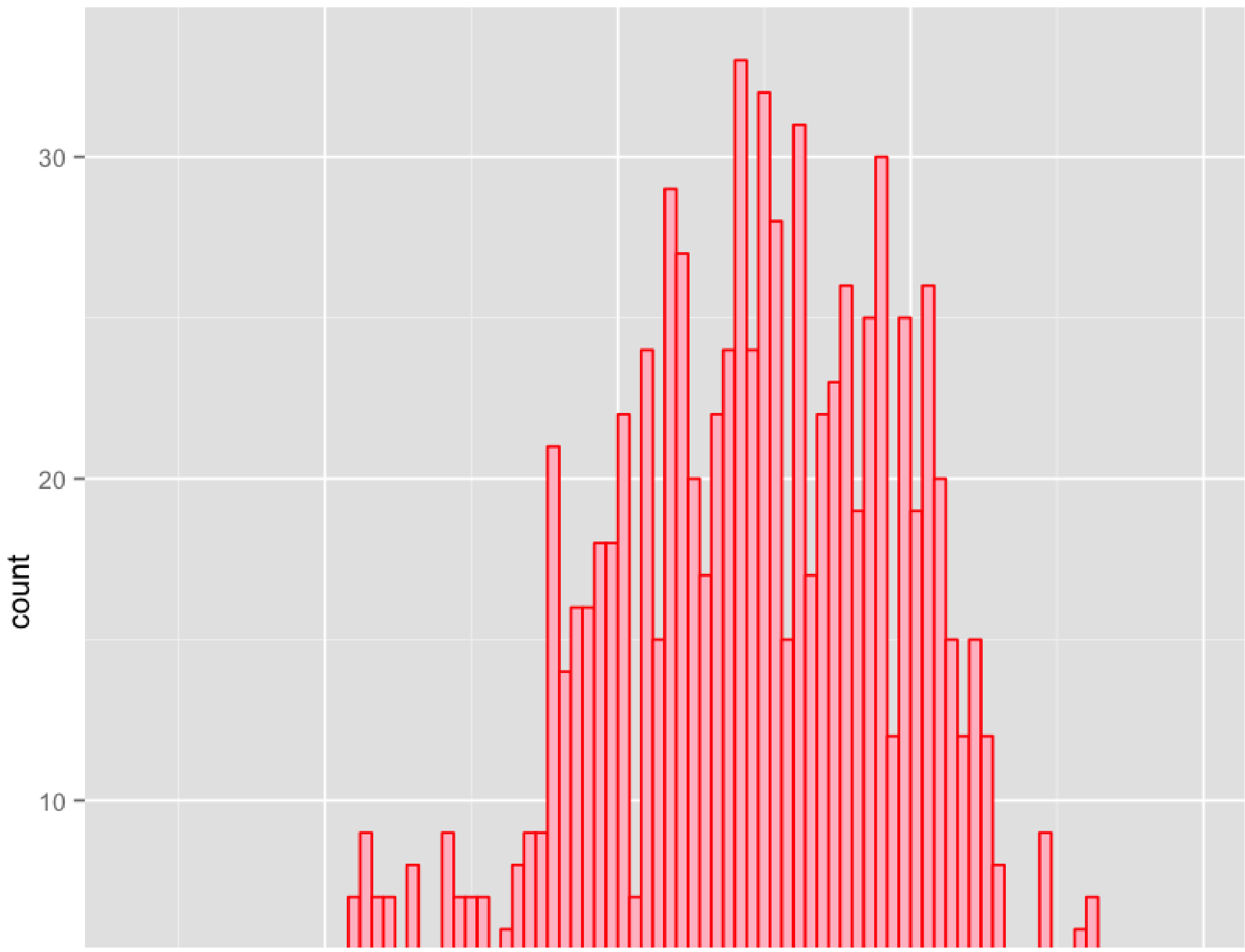


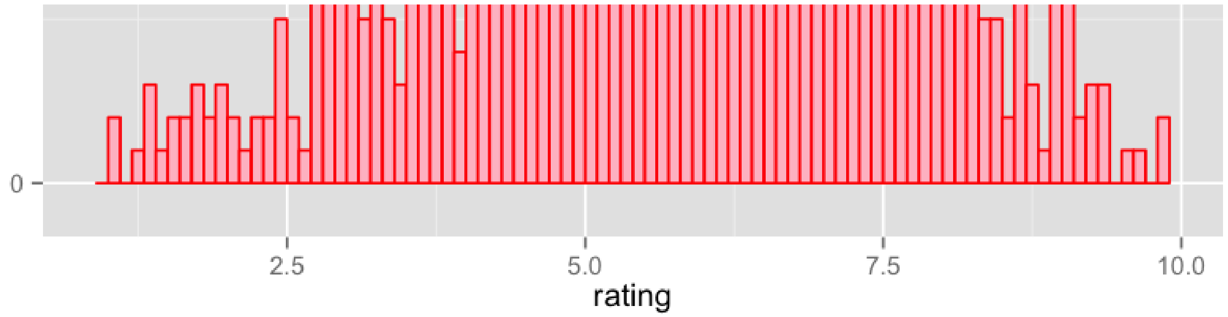


Adding Color

In [41]:

```
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(binwidth=0.1,color='red',fill='pink')
```

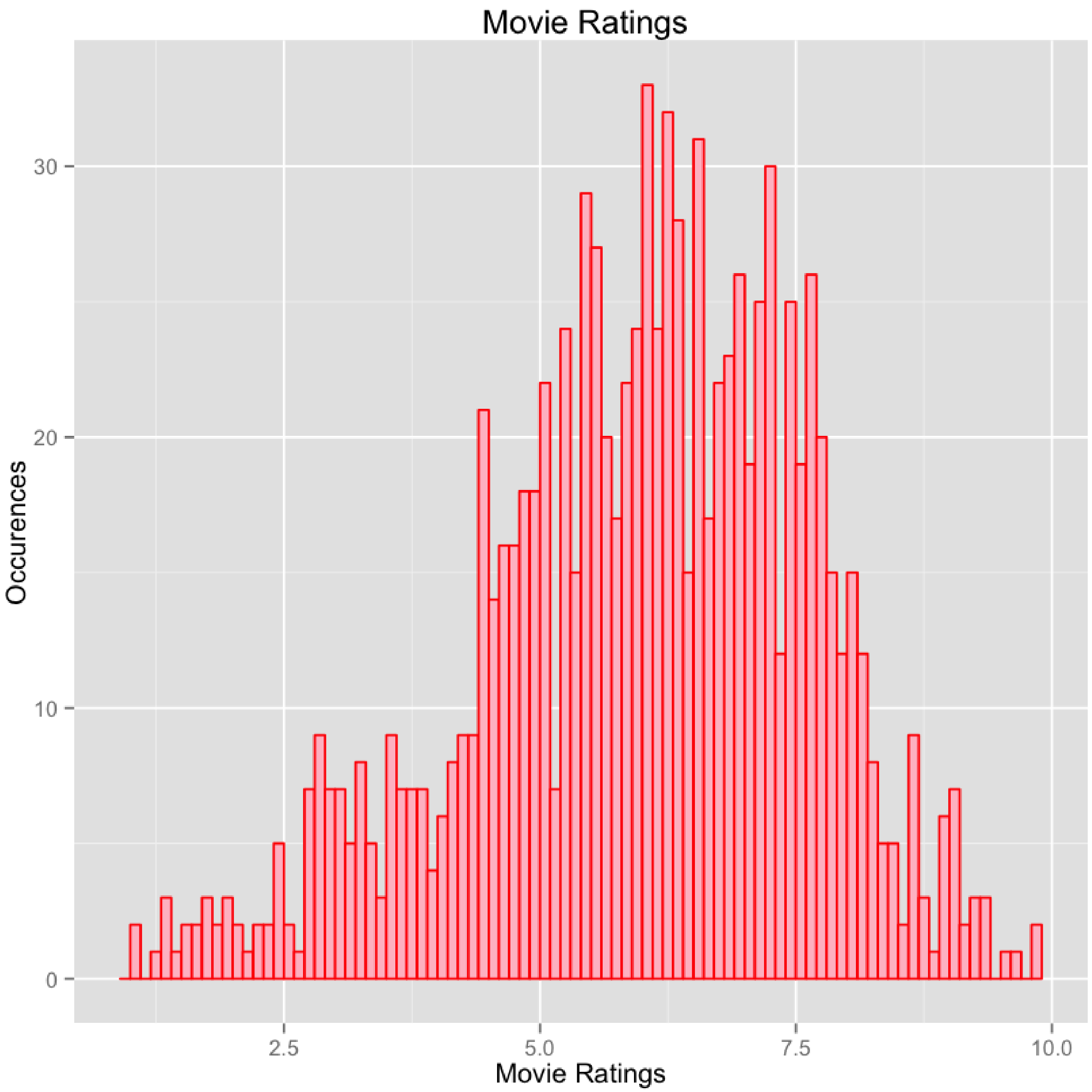




Adding Labels

In [65]:

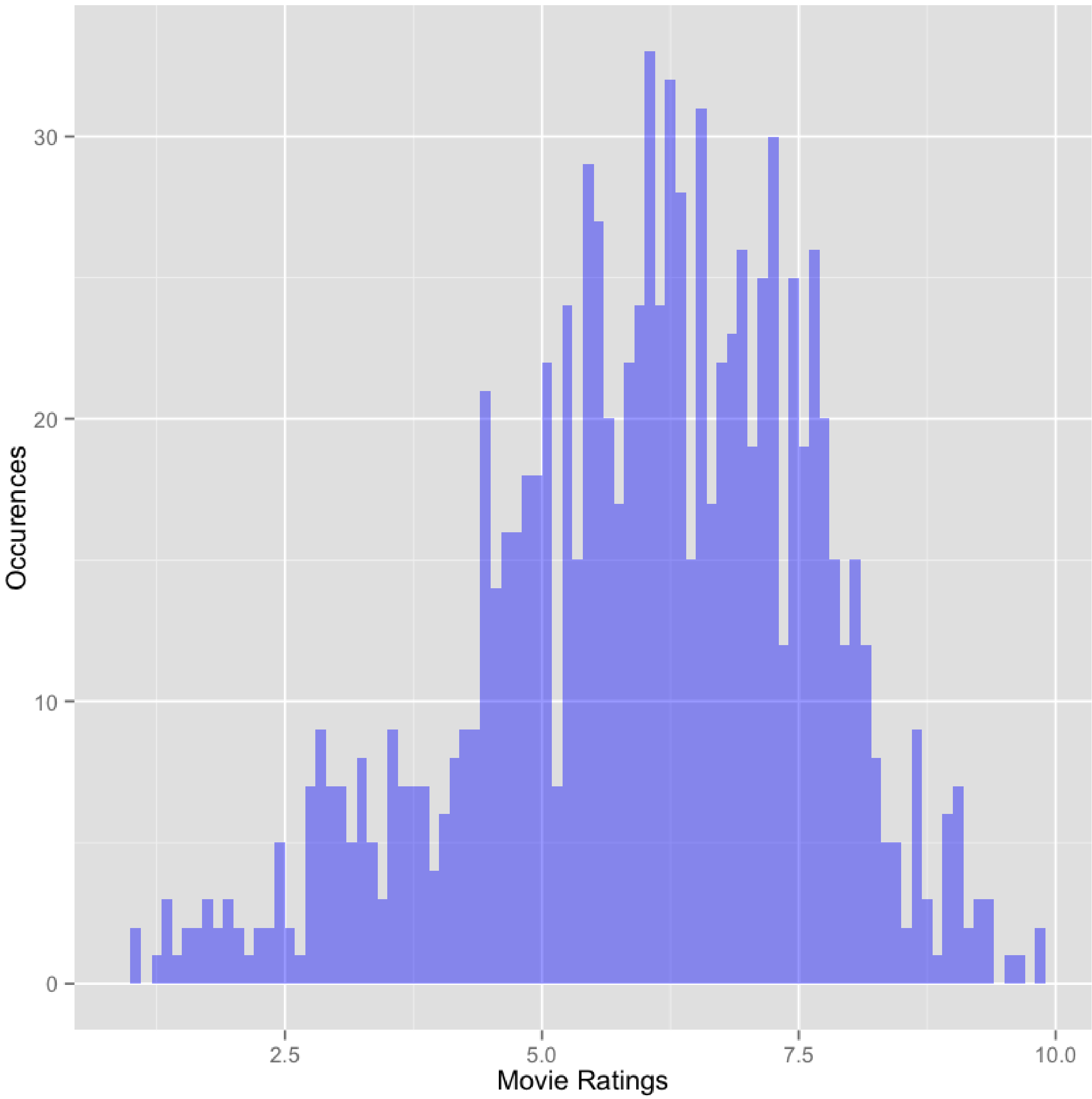
```
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(binwidth=0.1,color='red',fill='pink') + xlab('Movie Ratings')+ ylab('Occurences') +
ggtitle(' Movie Ratings')
```



Change Alpha (Transparency)

In [49]:

```
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(binwidth=0.1,fill='blue',alpha=0.4) + xlab('Movie Ratings')+ ylab('Occurences')
```

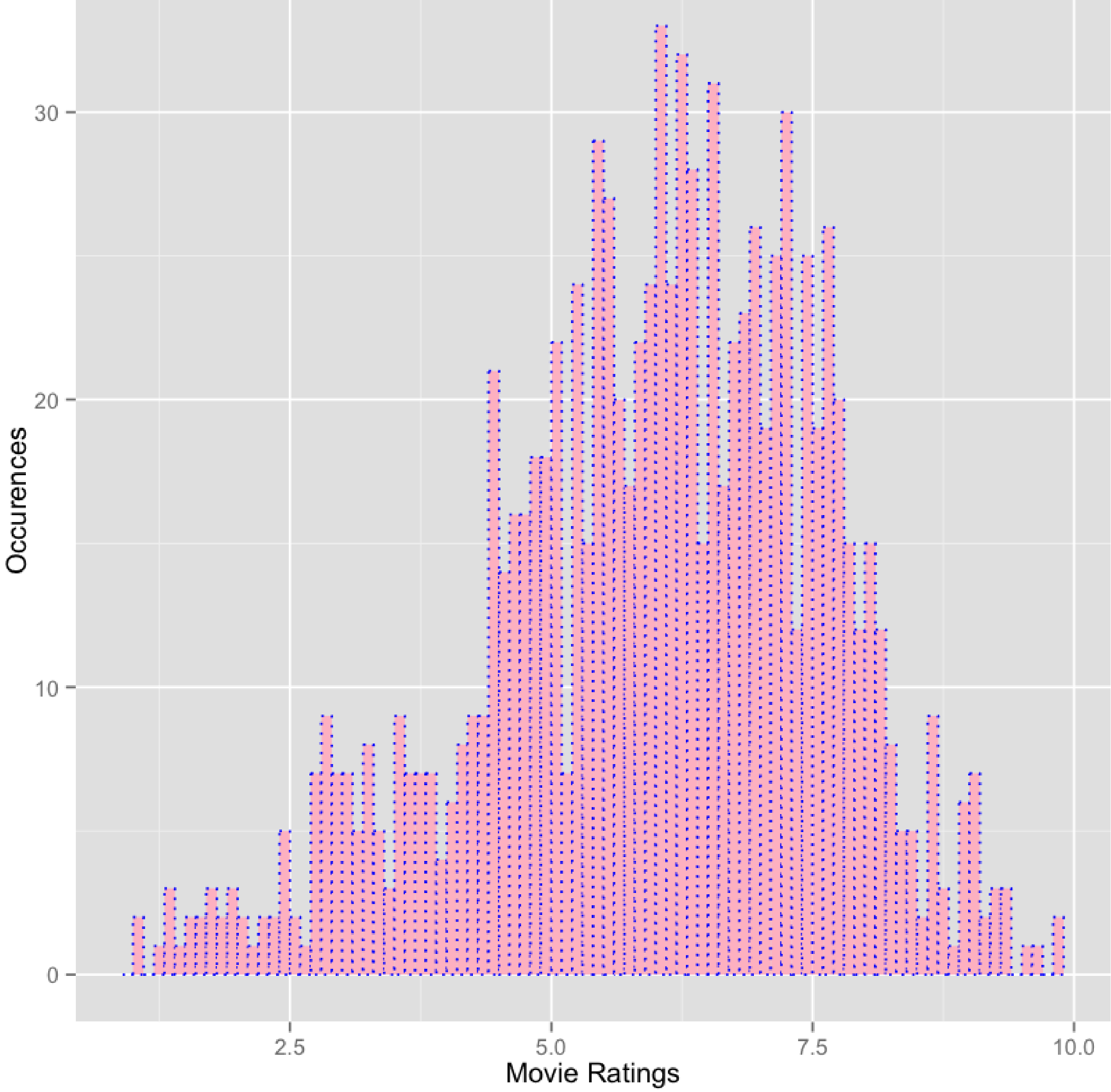


Linetypes

We have the options: "blank", "solid", "dashed", "dotted", "dotdash", "longdash", and "twodash". You would never really use these with a histogram, but just to show your options:

In [52]:

```
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(binwidth=0.1,color='blue',fill='pink',linetype='dotted') + xlab('Movie Ratings')+ y
lab('Occurences')
```

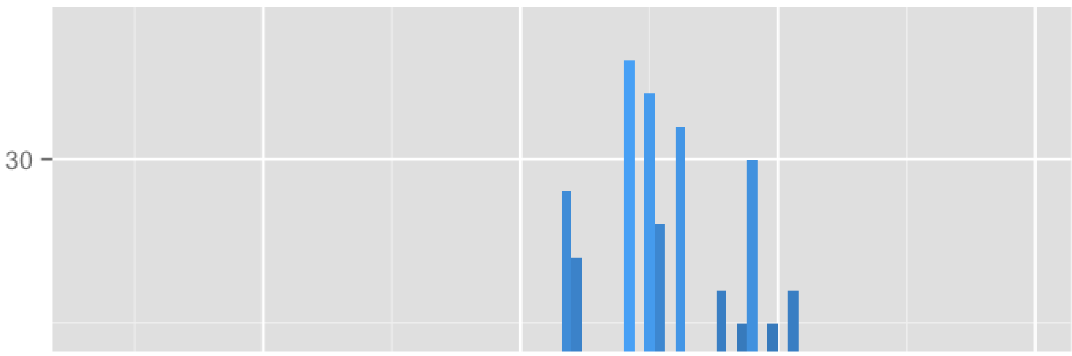


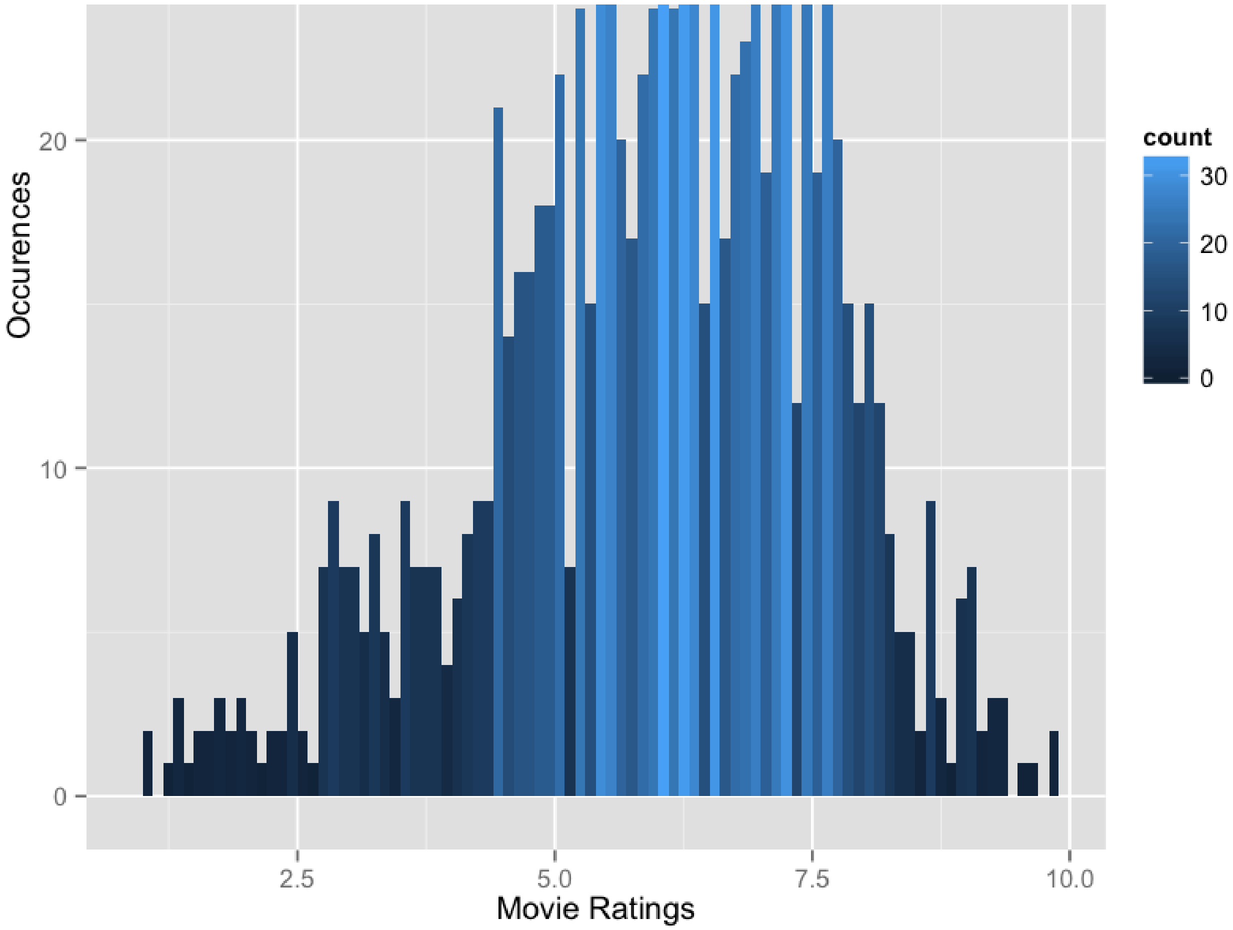
Advanced Aesthetics

We can add a **aes()** argument to the geom_histogram for some more advanced features. We won't go too deep into these, but ggplot gives you the ability to edit color and fill scales.

In [57]:

```
# Adding Labels
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(binwidth=0.1,aes(fill=..count..)) + xlab('Movie Ratings')+ ylab('Occurences')
```





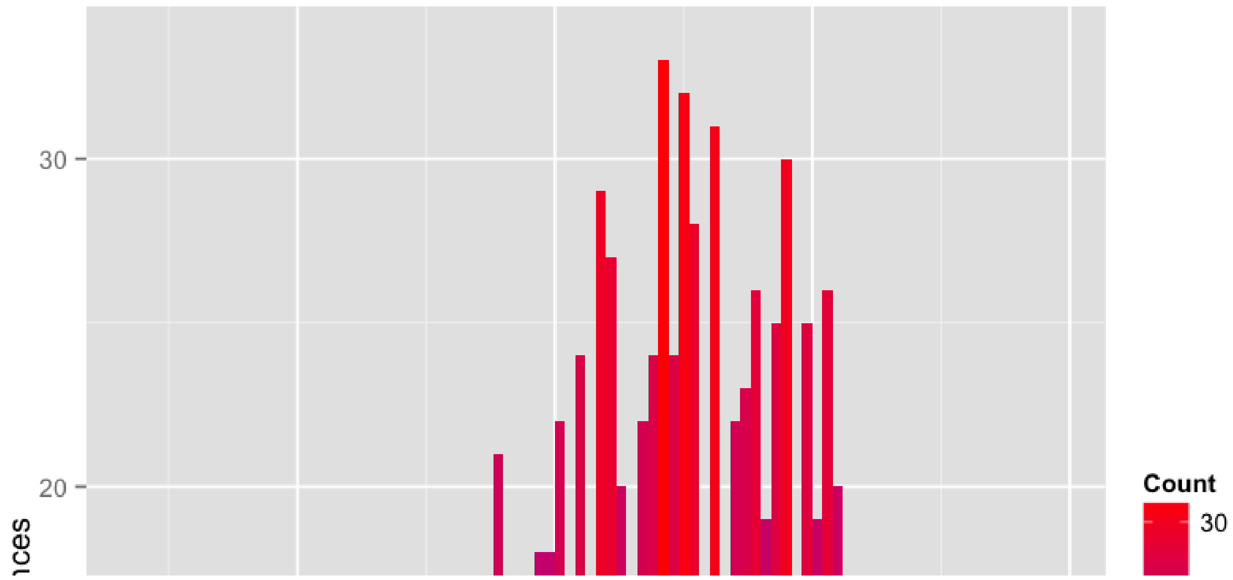
You can further edit this by adding the **scale_fill_gradient()** function to your ggplot objects:

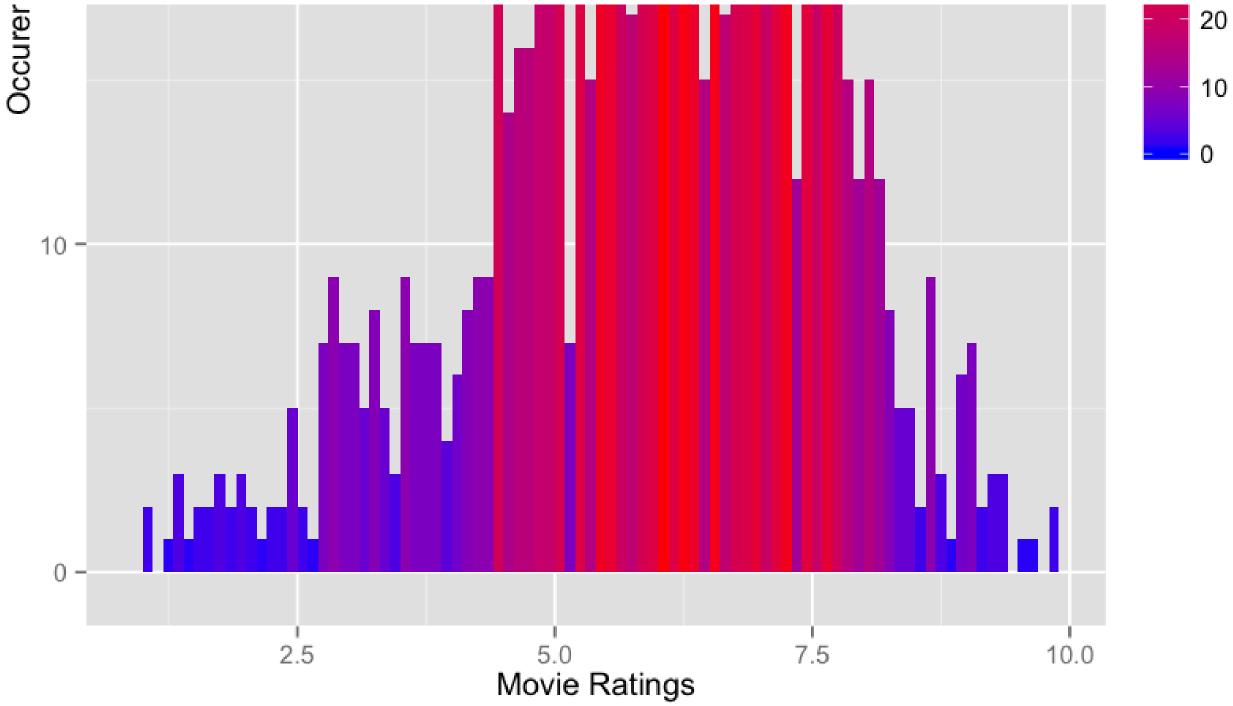
In [63]:

```
# Adding Labels
pl <- ggplot(df,aes(x=rating))
pl2 <- pl + geom_histogram(binwidth=0.1,aes(fill=..count..)) + xlab('Movie Ratings')+ ylab('Occurences'
)
```

In [59]:

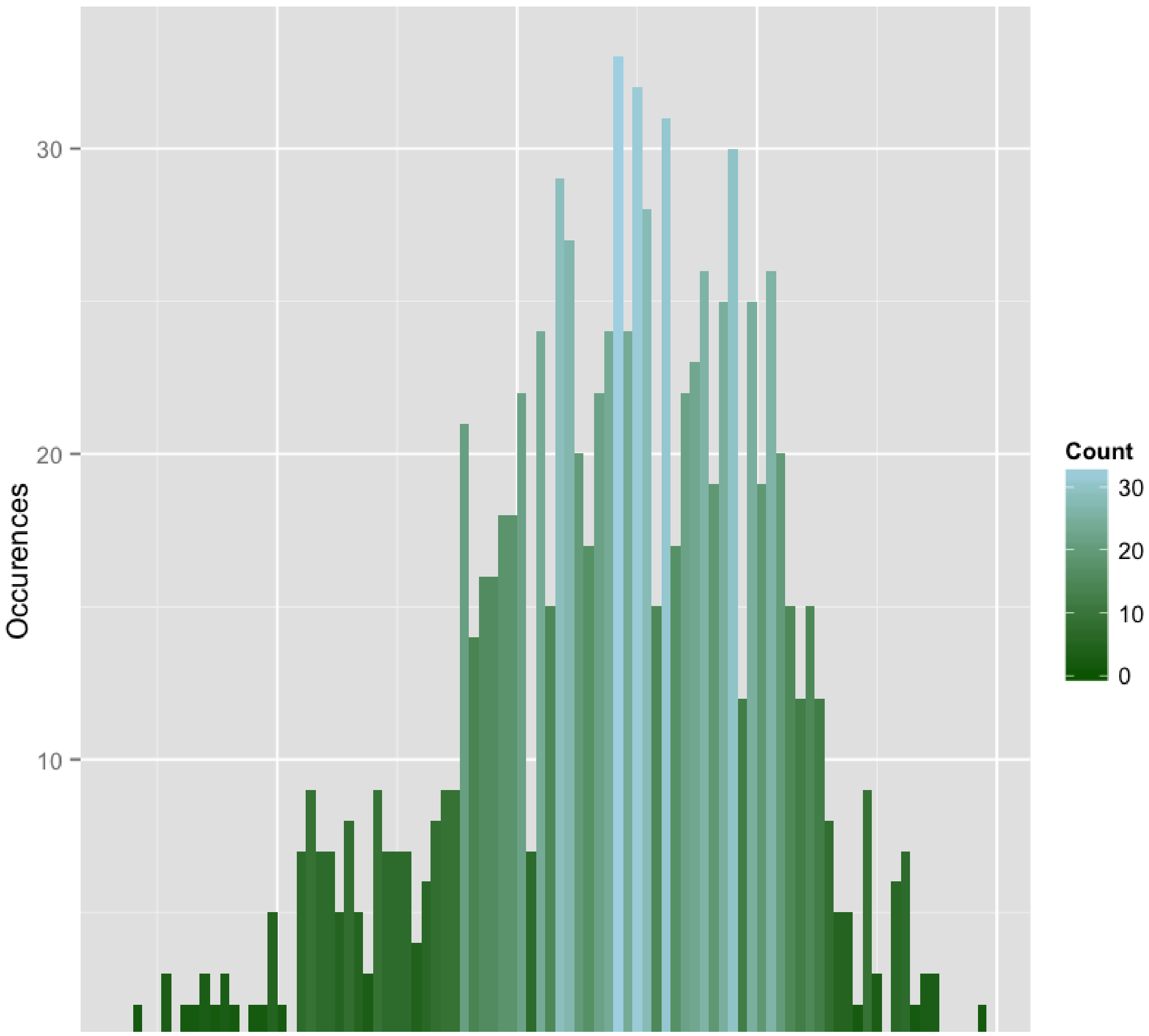
```
# scale_fill_gradient('Label',low=color1,high=color2)
pl2 + scale_fill_gradient('Count',low='blue',high='red')
```

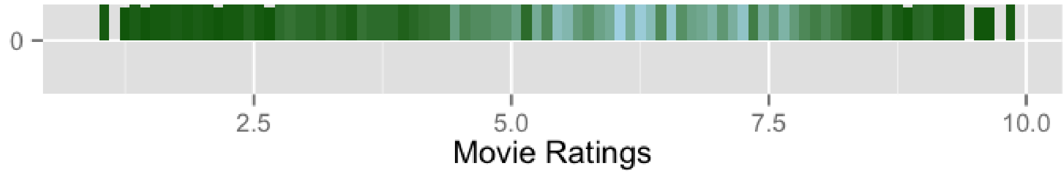




In [62]:

```
# scale_fill_gradient('Label',low=color1,high=color2)
pl2 + scale_fill_gradient('Count',low='darkgreen',high='lightblue')
```





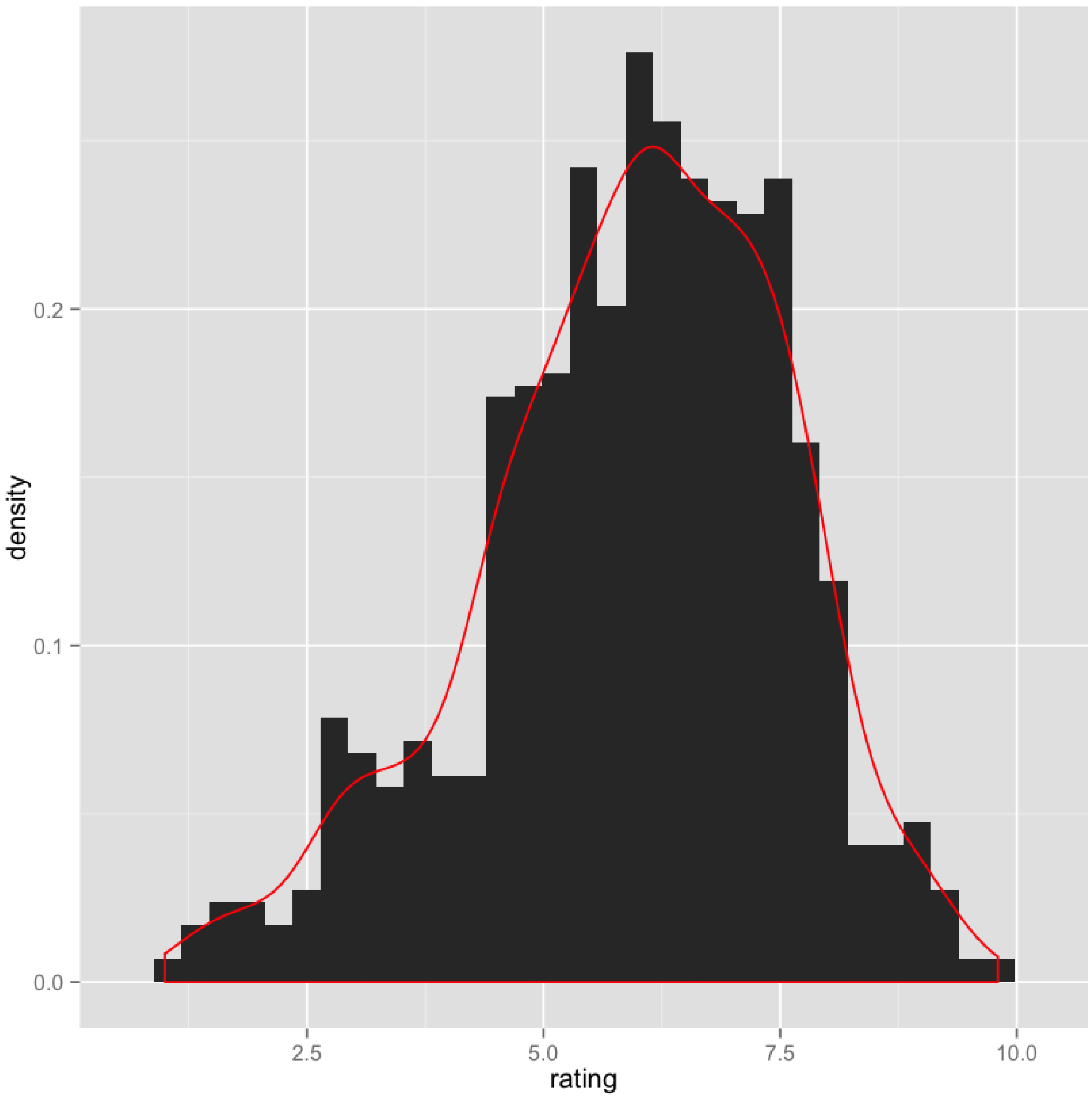
Adding density plot

You can add a [kernel density estimation plot](https://en.wikipedia.org/wiki/Kernel_density_estimation)

In [68]:

```
# Adding Labels
pl <- ggplot(df,aes(x=rating))
pl + geom_histogram(aes(y=..density..)) + geom_density(color='red')
```

```
stat_bin: binwidth defaulted to range/30. Use 'binwidth = x' to adjust this.
```



Alright! That's all for now concerning histograms. We've shown that ggplot has amazing customization capabilities, however it definitely takes time to get used to!